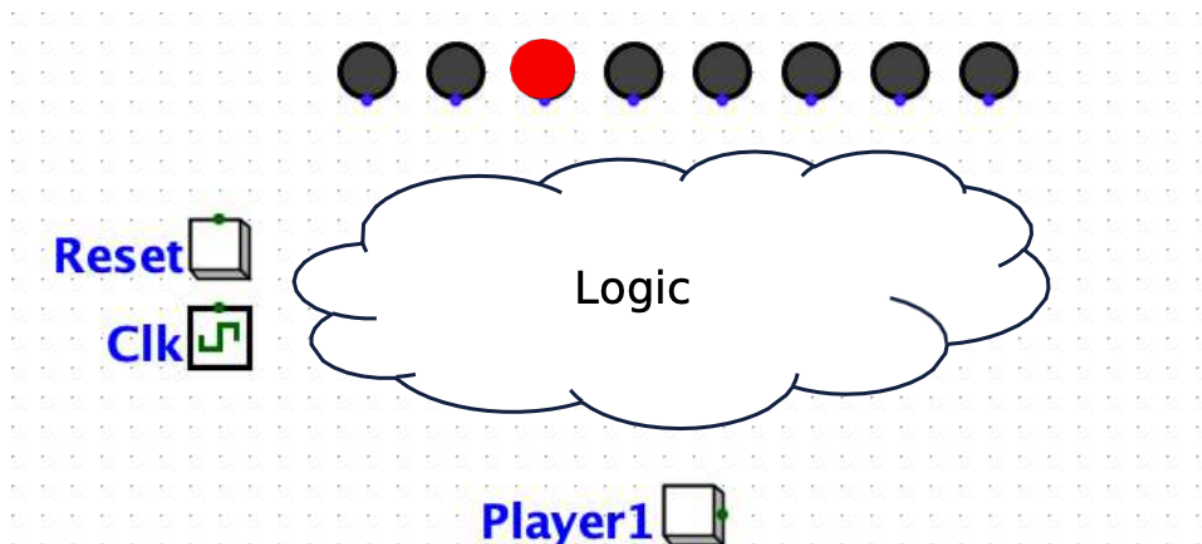


Implementation of Display Board, Lab Report (1st part)

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Digital Systems Laboratory



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1. Introduction

The objective of this second laboratory session is the hardware implementation and validation of the Display subsystem of a digital system designed to simulate the movement of a ball. This module serves as the primary visual interface, translating binary position data into a sequential LED display. As we did with the clock circuit, we followed the strategy of initially implementing the circuit on the breadboard to test its operational characteristics and then in the next lab report the full soldered display-clock circuit is

going to be presented on the report. The main focus of this report is to present and highlight the decoder + Leds+ resistor circuit implementation and logic.

The hardware design is based on standard TTL (Transistor-Transistor Logic) integrated circuits. We utilized a 74191 4-bit up/down counter to track position and a 74155 decoder to drive the individual LED lamps. A key challenge in this implementation was managing the negative logic characteristic of TTL decoders. Since the 74155 operates by pulling only one output to a logical Low (0) while keeping the others High (1), the LEDs had to be wired in a specific configuration to light up correctly.

Beyond the display logic, the board also integrates the primary user interfaces, specifically the Reset and Play switches. These were implemented using a pull-up resistor configuration to ensure signal stability and define "Active Low" behavior for the control logic. This report details the interface specifications, the layout strategies used to keep the board organized, and the initial testing phases before the full control logic is added in the next lab.

2. Interface description

In this 1st part of the Pong Project's display system we focus on creating a fully operating decoder, leds subsystem with a tricky active low logic, powered by a 5V source. The breadboard implementation is presented below:

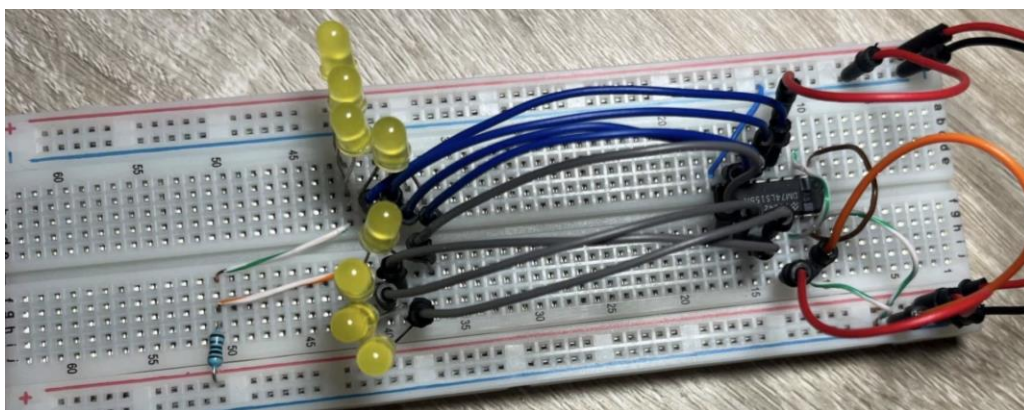


Figure 1: Prototype of the decoder-leds-resistor display subsystem on a breadboard

The "Display" board serves as the bridge between the future control logic and the user. With the addition of the counter + clock circuits a complete signal interface array is going to be presented and analyzed to ensure compatibility and well described circuit behavior. For this stage the signals coming in to the decoder and the leds are presented in the following smaller signal array.

Interface Table of the Decoder + LEDs Subsystem

Signal Name	Direction	Width	Description	Polarity
Vdd	Input	1 bit	+5 V supply of the subsystem.	-
Gnd	Input	1 bit	Ground reference of the subsystem.	-
Sel2	Input	1 bit	Most significant bit of the 3-bit selection input of the 74155.	Active high
Sel1	Input	1 bit	Middle bit of the 3-bit selection input of the 74155.	Active high
Sel0	Input	1 bit	Least significant bit of the 3-bit selection input of the 74155.	Active high
En	Input	1 bit	Decoder enable signal.	Active low
LED[7:0]	Output (visual)	8 bit	Visual representation of the decoded position by means of 8 LEDs.	Activated through the active-low outputs of the 74155

Note: The signals Sel2, Sel1, and Sel0 correspond to the three binary selection inputs of the 74155 decoder in the present subsystem. Their interconnection with the 74191 counter belongs to the next implementation stage of the complete Display subsystem.

3 - Circuit implementation

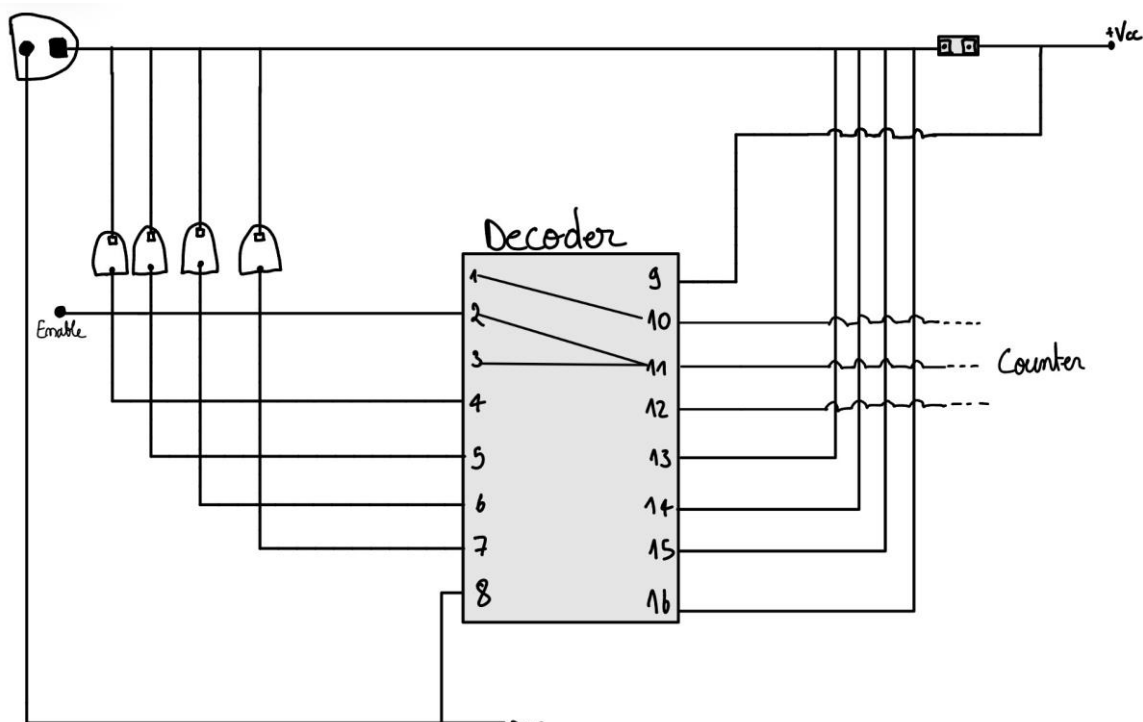
This part of the report is dedicated to the actual logic design of the display's decoding subsystem and the technical attributes of it. We are going to analyze how he managed to use the TTL's pins in order to turn it into the 3->8 decoder we need and then use its active-low logic to power only one LED at a time.

More specifically, half of the display logic relies on the 74155 decoder, which translates the 3-bit binary output from the counter(more details about the counter will be given in the next lab report) into a single active signal. Because TTL components frequently implement negative logic, the decoder operates by outputting a logical High (1) on N-1 pins, and a logical Low (0) exclusively on the selected output pin. Consequently, the 8 LEDs are wired so that they only illuminate when the decoder output is driven Low.

To achieve this, the LEDs are connected between the +5V supply and the decoder outputs, passing through a current-limiting resistor. A key design choice is the use of a single 1kΩ resistor for all 8 LEDs. Since normal operation dictates that at most one LED is active at any given time, a single resistor is sufficient. If a malfunction were to activate multiple outputs simultaneously, the shared resistor would safely limit the total current, preventing hardware damage while resulting in a noticeable decrease in LED brightness. Additionally, a separate red-colored LED - with its own resistor connected directly to the

power supply – will be added in the next full display report to serve as a constant power indicator.

For user input, the board will feature two push-button switches: Play and Reset. These will be wired in a configuration where the switch pulls the output to 0 (ground) when pressed, and a pull-up resistor pulls the output to 1 when open. The necessary resistor will be added to prevent a short-circuit when the switch is closed. As a consequence of this specific wiring, both the Reset and Play control signals are going to be "Active Low".



Decoder configuration and internal pin logic

In more detail, although the 74155 is internally implemented as a **dual 2-to-4 decoder/demultiplexer**, in the present subsystem it is configured so as to behave as a **3-to-8 decoder**, which is exactly what the Pong display requires. This is achieved by using two address inputs for local selection and one additional input for section selection. According to the decoder logic, the outputs are active low, the address lines **A** and **B** are common to both internal sections, while the two sections are enabled under different conditions: section 1 is enabled when **1G = 0** and **1C = 1**, whereas section 2 is enabled when **2G = 0** and **2C = 0**.

In the complete display implementation, the three binary bits that encode the ball position are provided by the counter. In the present report, however, these three bits are

treated simply as the three binary selection inputs of the decoder subsystem. Their practical use inside the 74155 can be summarized as follows:

- **Pin 13 (A)** receives the least significant selection bit.
- **Pin 3 (B)** receives the second selection bit.
- **Pins 1 (1C) and 15 (2C)** are tied together and receive the third selection bit, which determines **which of the two internal 2-to-4 sections becomes active**.
- **Pins 2 (1G) and 14 (2G)** are also tied together and form a **common enable line** for the decoder.

This connection scheme allows the device to operate as an 8-output decoder:

- when the third input bit is **0**, the second internal section is enabled,
- when the third input bit is **1**, the first internal section is enabled,
- while the pair **A, B** selects one of the four outputs within the active section.

Therefore, the 8 LED-driving outputs of the 74155 are used as a single ordered set:

- **pins 9, 10, 11, 12** correspond to one group of four decoded positions,
- **pins 7, 6, 5, 4** correspond to the second group of four decoded positions.

Because only one internal section is active at a time, and because only one output of that section becomes LOW, the final result is that **exactly one LED is selected at any given moment**. This is the essential decoding behavior required for the visual representation of the ball position in the Pong display.

A particularly important control pin group is the decoder enable:

- **pins 2 and 14** are active-low enable inputs,
- when the common enable line is in the active state, decoding takes place normally,
- when the enable line is inactive, no valid decoded LED output is produced and the display can be forced into a **blank / no-LED state**. This is fully consistent with the lab requirement that the “goal” or inactive display condition can be implemented by exploiting the decoder enable mechanism.

Finally, the electrical LED wiring follows directly from the active-low nature of the decoder:

- the LEDs are connected between **+5 V** and the decoder outputs,

- the common supply branch includes a current-limiting resistor,
- thus, an LED lights up only when its corresponding decoder output is driven to **LOW**.

The official lab notes explicitly allow the use of a single resistor for all LEDs, since under correct operation only one LED is expected to be on at any given time.

4 - Circuit behavior and control strategy

The circuit's behavior was rigorously tested to ensure it accurately simulates the ball's movement. In the following report, the 74191 - which is acting as the position counter - will be included; with every incoming clock pulse (Clock_in), the counter increments or decrements based on the state of the UpDown directional signal. For now, we ensured that the 74155 decoder successfully translates this 3-bit value to drive exactly one of the 8 LEDs Low, creating the visual effect of a moving ball.

To implement the "goal/target achieved" state, we utilized the Enable (En) pin on the decoder. When En is rendered inactive, all decoder outputs are forced High, turning off all LEDs simultaneously. Furthermore, asserting the Core_reset signal successfully initializes the counter, returning the ball to its default starting position.

5 – Recording of execution/implementation experiences

The laboratory exercise provided valuable practical experience in transitioning from theoretical logic to physical hardware. Before soldering, the circuit was constructed at a logical level and tested on a breadboard to verify component behavior and confirm the decoder's negative logic functionality. Finally, this stage helped clarify the role of the decoder-LED circuit as a **partial implementation of the full display board**. Although the counter is not yet included in the present stage, the subsystem was designed with the next implementation step in mind, so that its three binary selection inputs can later be driven directly by the outputs of the 74191. In this sense, the present exercise was not only a construction task, but also a preparation stage for the complete display subsystem of the Pong project.

